

Nuclear Decay as a Spectral Bifurcation of the Flag Condensate: Instanton Action, Bogoliubov Mode Mixing, and Geometric Unification

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Abstract

We present a particle-free formulation of nuclear decay and quantum tunneling within the Algebraic Conductance State (ACS) flag-condensate ontology. By representing the nucleus as a confined standing-wave phase lock of canonically conjugate real field components ϕ (cosine mode) and π (sine mode), decay is reinterpreted as a spectral bifurcation into a travelling wave driven by geometric phase slips across the colour-confining throat. We derive the WKB Gamow transmission factor $T = e^{-2W}$ from the Euclidean deformation of the BPST instanton action $\delta S(R)$ on the curved throat metric. Utilizing a numerically stable 2×2 transfer matrix framework, we verify the Bogoliubov mode-mixing relation $|\beta/\alpha|^2 = e^{-2W}$ across 14 alpha-emitting isotopes (^{212}Po to ^{244}Cm) spanning many orders of magnitude in half-life. This yields a Geiger–Nuttall linearity of $R^2 = 0.9939$ with a predicted-to-measured slope ratio of 1.0000 on the stated set, documenting that the decay *slope* is geometric while the intercept encodes the alpha-cluster preformation amplitude extracted from data (P_α not predicted from first principles in this note). Applying the identical transfer matrix formulation to Schwarzschild horizons recovers the Hawking temperature $T_H = \hbar\kappa/(2\pi k_B)$ in the units used below. Nuclear decay, Hawking radiation, electroweak sphaleron transitions, and fluxon nucleation are presented as structurally parallel geometric limits of one phase-defect mechanism; absolute physical identity beyond that structural parallel is not claimed.

Publication status (24 September 2026). This version includes a claim-specific amendment at the end of the main text. That amendment governs earlier physical interpretations and distinguishes retained calculations from withdrawn or unresolved inferences.

1 Introduction

Standard nuclear physics models alpha decay as a preformed quantum particle tunneling through a classical Coulomb barrier [1, 2]. While computationally successful, this picture relies on the dual assumptions of point-like particle localization and ad-hoc barrier penetration probabilities.

In this work, we eliminate particle-based tunneling axioms entirely. Within the Algebraic Conductance State (ACS) framework, the complex scalar flag field $\Phi(\mathbf{r}, t)$ is the sole ontological primitive. What is conventionally termed an “alpha particle” is shown to be a propagating sine/cosine wave packet—a difference-frequency *Tartini tone* generated by non-linear mode mixing inside the colour-confining nuclear throat.

We present the complete granular mathematical derivation, numerical validation, and geometric cross-domain unification connecting nuclear decay to Hawking radiation and topological instantons.

2 The Flag Condensate Field

2.1 Phase Decomposition

Let the complex scalar field $\Phi(\mathbf{r}, t)$ represent the flag condensate amplitude:

$$\Phi(\mathbf{r}, t) = \frac{1}{\sqrt{2}} \left[\phi(\mathbf{r}, t) + i \pi(\mathbf{r}, t) \right], \quad (1)$$

where $\phi(\mathbf{r}, t)$ and $\pi(\mathbf{r}, t)$ are canonically conjugate real fields satisfying equal-time commutation relations:

$$[\phi(\mathbf{r}, t), \pi(\mathbf{r}', t)] = i\hbar \delta^{(3)}(\mathbf{r} - \mathbf{r}'). \quad (2)$$

2.2 Interior: Confined Standing-Wave Lock

Inside the nuclear throat ($r < R$), strong colour confinement binds the flag field into a stationary eigenstate:

$$\phi_{\text{in}}(r, \theta, \varphi, t) = A j_\ell(kr) Y_{\ell m}(\theta, \varphi) \cos(\omega t), \quad (3)$$

$$\pi_{\text{in}}(r, \theta, \varphi, t) = A j_\ell(kr) Y_{\ell m}(\theta, \varphi) \sin(\omega t), \quad (4)$$

where $j_\ell(kr)$ is the spherical Bessel function vanishing at boundary $r = R$, and $Y_{\ell m}$ are spherical harmonics. Because ϕ_{in} and π_{in} are strictly quadrature-locked in time (90° out of phase), spatial energy density remains localized:

$$\mathcal{E}_{\text{in}} = \frac{1}{2}(\dot{\phi}^2 + (\nabla\phi)^2 + \dot{\pi}^2 + (\nabla\pi)^2) = \text{const}. \quad (5)$$

No localized physical particle exists within the interior—only a stationary saturation node of the flag field.

2.3 Exterior: Free Travelling Wave

Outside the barrier ($r > b$), the field propagates as an outgoing spherical wave:

$$\phi_{\text{out}}(r, \theta, \varphi, t) = \text{Re} \left[B h_\ell^{(1)}(kr) Y_{\ell m} e^{-i\omega t} \right], \quad (6)$$

$$\pi_{\text{out}}(r, \theta, \varphi, t) = \text{Im} \left[B h_\ell^{(1)}(kr) Y_{\ell m} e^{-i\omega t} \right], \quad (7)$$

where $h_\ell^{(1)}$ is the spherical Hankel function of the first kind. Here, ϕ and π propagate *together* in phase, representing a complete spectral bifurcation from standing-wave confinement to travelling-wave freedom.

3 Instanton Action on the Confining Throat

3.1 Euclidean Deformation

In flat space $\mathbb{R}^4$, the classical Yang-Mills BPST instanton action [3] is a pure topological invariant:

$$S_0 = \frac{8\pi^2}{g^2}, \quad (8)$$

with g the gauge coupling of the confining sector under consideration. However, the colour-confining throat radius R introduces a physical scale that breaks conformal invariance. The metric of the effective throat takes an AdS-like warped geometry:

$$ds^2 = \frac{R^2}{r^2} dr^2 + \frac{r^2}{R^2} \eta_{\mu\nu} dx^\mu dx^\nu. \quad (9)$$

Evaluating the Euclidean action on this background gives the deformed action $\delta S(R)$ (the dependence on the Coulomb barrier V_C enters through κ):

$$\begin{aligned} S_E &= S_0 + \delta S(R) \\ &= \frac{8\pi^2}{g^2} + 2 \int_R^b \kappa(r) dr, \end{aligned} \quad (10)$$

where $\kappa(r) = \frac{\sqrt{2\mu(V(r)-E)}}{\hbar}$ is the wave suppression parameter. In the semiclassical limit where throat radius $R \ll b$, the geometric deformation term $\delta S(R) \gg S_0$ dominates the action penalty.

3.2 The Gamow Phase Defect

The Gamow integral W measures the cumulative geometric phase mismatch:

$$W \equiv \int_R^b \kappa(r) dr = \frac{\sqrt{2\mu}}{\hbar} \int_R^b \sqrt{\frac{k_e Z_1 Z_2 e^2}{r} - E} dr. \quad (11)$$

Integration yields the exact analytic expression:

$$W = \eta_S \left[\arccos(\sqrt{R/b}) - \sqrt{\frac{R}{b} \left(1 - \frac{R}{b}\right)} \right], \quad (12)$$

where $\eta_S = \frac{k_e Z_1 Z_2 e^2}{\hbar v}$ is the Sommerfeld parameter and $v = \sqrt{2E/\mu}$ is the relative velocity.

4 Transfer Matrix Bogoliubov Extraction

4.1 Transfer Matrix Method

To extract transmission without numerical overflow at extreme ratios ($T \sim 10^{-40}$), we slice the potential barrier $[R, b]$ into N thin layers of width $\Delta r = (b - R)/N$. Each layer i possesses a 2×2 transfer matrix M_i :

For classically forbidden regions ($V > E$):

$$M_i = \begin{pmatrix} \cosh(\kappa_i \Delta r) & \frac{1}{\kappa_i} \sinh(\kappa_i \Delta r) \\ \kappa_i \sinh(\kappa_i \Delta r) & \cosh(\kappa_i \Delta r) \end{pmatrix}. \quad (13)$$

For allowed regions ($V < E$):

$$M_i = \begin{pmatrix} \cos(k_i \Delta r) & \frac{1}{k_i} \sin(k_i \Delta r) \\ -k_i \sin(k_i \Delta r) & \cos(k_i \Delta r) \end{pmatrix}. \quad (14)$$

The total barrier transfer matrix $M = \prod_{i=1}^N M_i$ is real and satisfies $\det(M) = 1.0$. The transfer-matrix transmission coefficient T is computed from matrix elements:

$$T = \frac{4k_{\text{in}}k_{\text{out}}}{(k_{\text{out}}M_{11} + k_{\text{in}}M_{22})^2 + (k_{\text{in}}k_{\text{out}}M_{12} - M_{21})^2}. \quad (15)$$

4.2 Bogoliubov Transformation

In mode space, the barrier acts as a Bogoliubov mixer relating asymptotic creation/annihilation operators:

$$b_k = \alpha_k a_k + \beta_k a_{-k}^\dagger, \quad |\alpha_k|^2 - |\beta_k|^2 = 1. \quad (16)$$

The transmission coefficient equals the squared mode-mixing ratio:

$$T = \left| \frac{\beta_k}{\alpha_k} \right|^2 = e^{-2W}. \quad (17)$$

Thus, the decay probability per assault cycle $\nu = v/(2R)$ gives the model half-life (with P_α data-extracted, per the Remark above):

$$\lambda = \nu \cdot P_\alpha \cdot e^{-2W} \implies t_{1/2} = \frac{\ln 2}{\lambda}, \quad (18)$$

where P_α is the standing-wave mode overlap fraction for the alpha configuration.

5 Numerical Verification & Results

Table 1: **Unified Numerical Results for 14 Alpha-Emitting Nuclei.** Cross-validation of WKB analytic Gamow integral W_{ana} , numeric Simpson integration W_{num} , 2×2 Transfer Matrix extraction W_{TM} , predicted Gamow half-lives, measured half-lives, and extracted alpha preformation factors P_α .

Isotope	Q_α (MeV)	R (fm)	b (fm)	W_{ana}	$t_{1/2}^{\text{Gamow}}$	$t_{1/2}^{\text{meas}}$	$\log_{10} P_\alpha$
^{212}Po	8.954	8.99	26.33	14.821	8.74×10^{-9} s	2.99×10^{-7} s	-1.53
^{214}Po	7.833	9.02	30.10	18.204	1.05×10^{-5} s	1.64×10^{-4} s	-1.19
^{216}Po	6.906	9.05	34.14	21.720	0.0132 s	0.145 s	-1.04
^{218}Po	6.115	9.07	38.56	25.549	28.45 s	186.0 s	-0.82
^{220}Rn	6.405	9.10	37.71	24.629	4.18 s	55.6 s	-1.12
^{224}Ra	5.789	9.16	42.67	28.514	9.21×10^3 s	3.16×10^5 s	-1.54
^{226}Ra	4.871	9.18	50.70	34.341	9.45×10^8 s	5.05×10^{10} s	-1.73
^{228}Th	5.520	9.21	45.72	30.829	8.07×10^5 s	6.04×10^7 s	-1.87
^{232}Th	4.082	9.26	61.84	42.673	2.98×10^{15} s	4.43×10^{17} s	-2.17
^{234}U	4.858	9.29	52.92	35.830	1.76×10^{10} s	7.74×10^{12} s	-2.64
^{238}U	4.270	9.34	60.20	41.517	1.52×10^{15} s	1.41×10^{17} s	-1.97
^{238}Pu	5.593	9.34	45.98	31.066	1.44×10^6 s	2.77×10^9 s	-3.28
^{240}Pu	5.256	9.36	48.93	33.626	2.55×10^8 s	2.07×10^{11} s	-2.91
^{244}Cm	5.902	9.42	44.47	29.832	1.39×10^5 s	5.72×10^8 s	-3.61

5.1 Geiger-Nuttall Law Linearity

Following the classical Geiger–Nuttall systematics [5], linear regression of $\ln \lambda$ against $Z/\sqrt{E}$ yields:

$$\text{Measured Fit: } \ln \lambda = -1.6842 \frac{Z}{\sqrt{E}} + 53.21, \quad R^2 = 0.9939, \quad (19)$$

$$\text{Predicted Fit: } \ln \lambda = -1.6841 \frac{Z}{\sqrt{E}} + 49.85, \quad R^2 = 0.9957. \quad (20)$$

The ratio of predicted to measured slopes is:

$$\frac{\text{Slope}_{\text{predicted}}}{\text{Slope}_{\text{measured}}} = 1.0000. \quad (21)$$

This confirms that the slope of the Geiger–Nuttall law is geometric (determined by throat potential curvature) on the stated isotope set, while the vertical offset encodes nuclear-structure content via the extracted preformation amplitude $\langle \log_{10} P_\alpha \rangle = -2.03 \pm 0.85$.

Remark (Status of P_α). P_α is *extracted* from the intercept (and from $t_{1/2}^{\text{meas}}/t_{1/2}^{\text{Gamow}}$), not predicted from first principles in this note. That is intentional modelling: the shape of the decay systematics is fixed by geometry; the absolute scale carries many-body nuclear physics. The standing-wave overlap sequel computes a geometric/structural preformation proxy $P_{\text{model}} = |\langle u_{\text{in}} | u_\alpha \rangle|^2$ on $[0, R]$ (optional global spectroscopic factor S) in [8]; absolute scale may still need S , and many-body uniqueness is not claimed.

6 Four-Domain Phase-Defect Pattern

Applying the identical transfer matrix / Bogoliubov formulation across four domains exhibits a shared structural pattern (phase defect W , mode mixing, exponential rate). This is a unification of *mechanism shape*, not a claim that the four phenomena are physically identical beyond that pattern.

For black holes, replacing the colour throat with the Schwarzschild horizon and setting surface gravity $\kappa = c^4/(4GM)$ evaluates the mode-mixing spectrum to the Hawking form [4]:

$$n(\omega) = \frac{|\beta_\omega|^2}{|\alpha_\omega|^2 - |\beta_\omega|^2} = \frac{1}{e^{2\pi\omega/\kappa} - 1}, \quad (22)$$

in units with $c = 1$. Equating this to a Planckian spectrum yields

$$T_H = \frac{\hbar\kappa}{2\pi k_B} = \frac{\hbar c^3}{8\pi G M k_B} \quad (23)$$

when c is restored.

Table 2: **Four-Domain Phase-Defect Unification.**

Domain	Phase Defect W	Observable
Nuclear Decay (^{238}U)	41.517	$t_{1/2} = 4.47 \times 10^9$ yr
Black Hole ($1M_{\odot}$)	0.000	$T_H = 6.17 \times 10^{-8}$ K
Sphaleron (Electroweak) [†]	~ 90	$B + L$ Violation Rate
Superconductor (NbTi) [†]	~ 17.4	Fluxon Nucleation Rate

[†]Illustrative order-of-magnitude entries from standard literature estimates; unlike the nuclear column they are not derived or cross-validated in this note’s code.

7 Conclusion

We have shown that alpha radioactivity can be formulated without particle-based tunneling postulates inside the flag-condensate ontology. Nuclear decay is a localized spectral bifurcation of $\Phi = (\phi + i\pi)/\sqrt{2}$ from confined standing wave to propagating travelling wave. The Gamow factor W is the geometric action deformation of the BPST instanton on the colour-confining throat. On the stated isotope set, the Geiger–Nuttall *slope* matches geometry while P_{α} remains an intercept-extracted nuclear-structure factor. Parallel transfer-matrix structure across nuclear decay, Hawking radiation, electroweak sphalerons, and superconducting fluxons supports a shared phase-defect pattern; companion electron geometry is developed in [6], and the broader ontology in [7].

Amendment: scope and evidence, 24 September 2026

The reported 14-isotope slope comparison uses measured decay inputs and does not predict preformation from first principles. A barrier factor, an assault rate and an interior channel overlap are separate ingredients. The later physical frontier retains counterexamples in which spectral information does not fix the overlap. The scalar binding study likewise has a specified charge and free-wave threshold, not a derivation of nuclear isotope lifetimes. Structural parallels to other tunneling problems must preserve these distinct physical inputs.

For the complete-action and microscopic-carrier audit, see the companion papers *Complete Scalar Actions and Identifiability Boundaries in the ACS Framework* and *Condensate Binding, Charge Protection, and the Boundary of Particle Identification* in this repository’s `papers/updates/` directory. The dated publication record separates fresh portable checks from preserved archival results; it does not claim a complete rerun of every historical experiment or a proof of RH.

References

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